The Essence of Life: A Thermodynamic Perspective

-speculative framework

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Abstract: Thermodynamically, life is a self-optimizing engine whose main "objective function" is the maximization of cumulative entropy production. Organisms archive historically successful, multi-step entropy-generating routes in two complementary media: heritable genes and rewritable experience. These routes are distilled into compact, modular "blueprints" that can be stored, recombined, and projected onto novel energetic landscapes. By detecting the fractal signatures embedded in environmental energy distributions, living systems rapidly retrieve and deploy entire libraries of these blueprints in parallel, thereby producing cumulative entropy gains far beyond the Markov horizon of single-step reactions. Evolutionary sophistication is measured by (i) the depth of forward planning, (ii) the acuity of fractal pattern recognition, and (iii) the fidelity with which high-yield entropy pathways are encoded and recalled. In humans, this optimization manifest as: externalized memory (text, code, cloud) and engineered hardware freeze the most productive entropy channels into durable artifacts—steam engines, power grids, search algorithms—turning civilization itself into a planet-scale, ever-expanding entropy-maximizing network.

Keywords:The essence of life;Cumulative Entropy Producer;Cumulative Entropy Planner; The Nature of Creature;

Introduction: Against the backdrop of thermodynamics, the direction of time is defined by the irreversible increase of entropy, and the evolution of the universe is fundamentally a continuous process unfolding from order to disorder. Yet within this trend toward increasing entropy, life exhibits a peculiar complexity and apparent purposefulness: it maintains the stability of its own structure while continuously consuming energy, dissipating heat, and accelerating the environment's progression toward greater disorder. Traditional perspectives, such as Schrödinger's [1], regard life as a "local reduction in entropy"—a kind of miracle seemingly resisting the Second Law. But what if we adopt a different perspective—that life does not defy entropy increase, but rather serves as one of its most efficient and sophisticated agents? Scholars such as Ilya Prigogine [2], James Lovelock [3], Eric Smith and Harold Morowitz [4], and Roderick C. Dewar [5] have discussed, from various angles, the role of life and ecosystems in the process of entropy production. The framework proposed in this paper closely aligns with their viewpoints.

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The following brief article discusses the role of living organisms in the thermodynamic process of entropy increase.

The Essence of Life:

Organisms record historically efficient multi-step high-entropy-output pathways through genetic and experiential mechanisms, encoding them into functional modules that are storable, composable, and extrapolatable. By performing fractal recognition and pattern matching on the topological structure of spatial energy distribution, these templates are selectively and batch-invoked to maximize cumulative entropy production across extended time scales. The more advanced the organism, the stronger its capacity for long-term sequential planning, the more refined its fractal identification of energy landscape topology, and the higher its efficiency in recording and encoding historical high-entropy-output pathways. Intelligent beings such as humans do not rely solely on passive genetic memory to store modularized pathways; instead, they extend this capability externally—through written records, evolving into electronic media storage, and further materializing these high-entropy-output pathways into tools and engineered devices that embody and automate their operation.

In my previous paper [6], we established a two-layer fiber bundle space model that incorporates the thermodynamic arrow of time and is compatible with both general relativity and quantum field theory. Within this framework, the probabilistic randomness in the Schrödinger equation arises because there is not just one, but multiple maximum-entropy paths available at the next moment.

Within this model, although equivalent maximum entropy paths may exist for the next moment, different choices among these equivalent paths lead to divergent evolutionary trajectories over extended time sequences. This divergence constitutes the essence of biological path selection. While immediate next-step choices may be thermodynamically equivalent, their long-term consequences differ significantly in terms of cumulative entropy production. Thus, biological decision-making is fundamentally a forward-looking process. Genetic memory and experiential memory equip organisms with the capacity to anticipate future path developments, going beyond mere Markovian decisions based solely on the present state.

For example, for an organism, placing wood either to the left by the river or to the right beside a fire might represent equally probable maximum entropy transitions in the immediate next step. However, their subsequent entropy production over time will differ substantially. Therefore, the organism performs a prospective evaluation—selecting, among several equivalent maximum entropy paths at each instant, the one that leads to the greatest cumulative entropy increase over multiple future steps. This evaluation relies on genetic and experiential memory, which essentially encode previously successful cumulative high-entropy-production pathways and extrapolate them into the future.

Even in the choice of whether or not to freeze food in a cold storage facility, the act of refrigeration—though seemingly entropy-reducing—can be understood from a broader thermodynamic perspective. When considering the local system as a whole, including both the refrigerator and its external environment, the thermodynamic evolution at the next moment may follow an equivalent maximum entropy production path compared to leaving the food unrefrigerated and allowing it to decay naturally. In other words, the instantaneous rate of entropy production may not differ significantly between the two choices. The refrigeration process itself consumes energy and releases waste heat, and the total entropy generated during this process is sufficient to ensure that the global entropy increase is no less than—and possibly even exceeds—that of natural decomposition. Thus, the maximum entropy paths at the immediate next moment are effectively equivalent.

However, the crucial difference emerges over multi-step temporal evolution and the resulting trajectory of cumulative entropy production. Although the two choices exhibit thermodynamic equivalence at the initial moment, the decision to refrigerate opens up a distinct future path topology. By delaying the inefficient breakdown of organic matter, refrigeration preserves food resources with high chemical potential energy, thereby enabling individuals or societies to sustain metabolic activity over longer timescales, engage in complex labor, operate advanced technologies, and drive organizational expansion. These subsequent activities—whether agricultural reproduction, industrial manufacturing, transportation, or information processing—are all high-entropy-producing processes.

This once again confirms a central insight: the decisions made by living organisms, and especially intelligent civilizations, are not based merely on Markovian choices that maximize instantaneous entropy production at each moment. Rather, they are guided by a forward-looking path evaluation mechanism that integrates historical experience and future modeling. What intelligent beings optimize is not just the immediate step, but the long-term cumulative entropy production, operating within the constraint of selecting among maximum entropy paths available at each successive moment.

Genetic and especially experiential memory are most effectively formalized through mathematical structures—that is, encoding experience via mathematical representations to ensure precise storage and retrieval.

Feature	Advantages of Mathematical Formalization
Compression	Abstracts vast amounts of concrete experience into concise rules (e.g., "fire heats objects")
Generalization	Applicable to novel situations (e.g., inferring combustibility of new materials)
Composability and Extensibility	Multiple rules can be combined to simulate complex behaviors (heat → steam → motion → tools)
Transmissibility	Easily shared across individuals (via language, symbols, education)

Feature	Advantages of Mathematical Formalization
Predictability	Enables forward simulation: logical chains such as "if A, then B"

The essence of science and technology in intelligent life is the encoding of past experiences that successfully opened cumulative high-speed entropy-increasing pathways into tools, machines, and code—transforming them into pre-established routes that do not need to be rediscovered. These modular structures enable intelligent agents to transcend the Markov limitation of making decisions based only on the current state for the next immediate step. Instead, they allow pursuit of maximal cumulative entropy production over multiple time steps—not resisting entropy increase, but actively guiding and amplifying it.

In the framework of SEQ network dynamics, technological behavior in intelligent organisms can be viewed as an optimized control process for maximizing cumulative entropy production. Given the current spatial energy distribution $\{E_i(t)\}$, the system does not simply select the path that maximizes instantaneous entropy rate $\dot{S}(t)$ in the next moment. Rather, drawing upon genetic and experiential memory, it activates a set of pre-encoded multi-step entropy-increasing path templates $\Gamma_k = \{\Delta S_k^{(1)}, \Delta S_k^{(2)}, ..., \Delta S_k^{(n)}\}$, and physically implements them through tools, devices, or algorithms.

These templates, once encapsulated programmatically and combined modularly, form stable high-entropy-production channels, significantly enhancing long-term efficiency of entropy flow. This establishes life as an active participant in cosmic evolution. From this perspective, living systems can be understood as units optimized for discovering and following paths of maximal cumulative entropy increase.

Human civilization as a whole can thus be seen as a large-scale network dedicated to discovering, optimizing, and solidifying such maximal cumulative entropy-increasing pathways:

- **Power systems** → enabling long chains from fossil fuels to electricity, mechanical work, and information processing;
- **The Internet** → accelerating knowledge diffusion, improving societal responsiveness and adaptive efficiency;
- **Space exploration** → probing new energy sources and novel topological configurations of space;

All of these continuously expand the available pathways for generating greater entropy, representing a collective, intentional alignment with the universe's fundamental tendency toward increasing disorder—not by opposing it, but by channeling it more effectively.

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